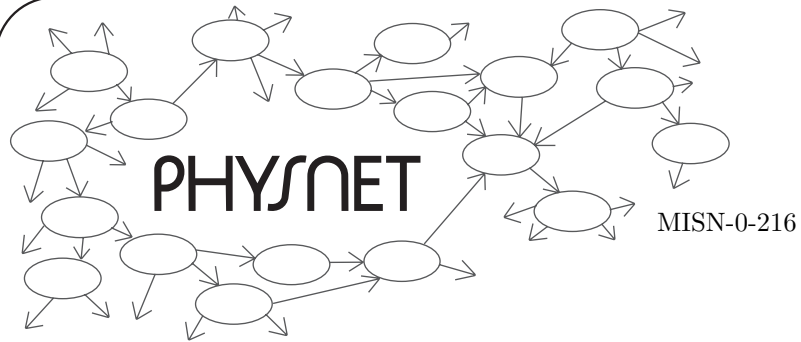
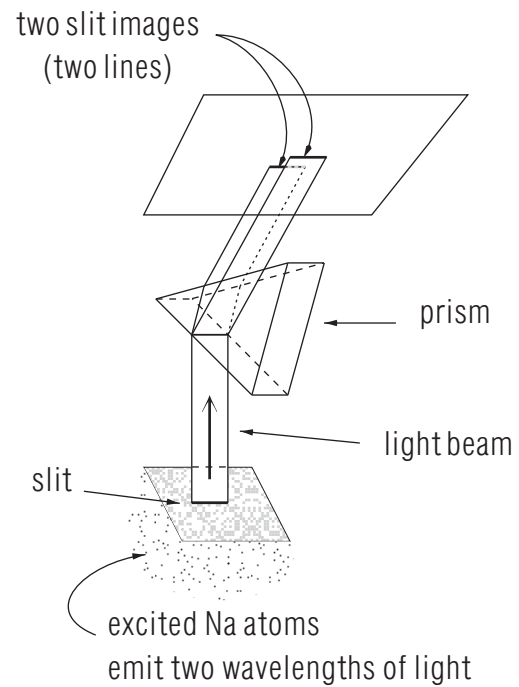




TRANSITIONS AND SPECTRAL ANALYSIS



TRANSITIONS AND SPECTRAL ANALYSIS

by
William C. Lane

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Title: **Transitions and Spectral Analysis**

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Evaluation: Stage B0

Length: 1 hr; 8 pages

Input Skills:

1. Vocabulary: atomic spectra, energy level, excited state, ground state (MISN-0-215).
2. Name the seven major regions of the electromagnetic spectrum (MISN-0-212).

Output Skills (Knowledge):

- K1. Vocabulary: band spectra, continuous and line spectra, fluorescence, phosphorescence, resolving power, dispersion, spectroscope.
- K2. State the three general types of information spectroscopy yields and cite an area of application for each type.
- K3. Compare and contrast emission spectroscopy with absorption spectroscopy, including the types of light sources used.
- K4. Briefly outline a procedure for spectral analysis, including: (a) the mechanism for exciting the sample atoms or molecules; (b) how the various wavelengths are separated; (c) why the spectra usually appear as lines; (d) how the wavelengths of the spectral lines are measured; and (e) how the sample atoms or molecules are then identified.
- K5. Describe the three mechanisms by which the atoms and molecules make transitions between energy levels.

External Resources (Required):

1. M. Alonso and E. J. Finn, *Physics*, Addison-Wesley (1970). For access, see this module's *Local Guide*.
2. I. M. Freeman, *Physics, Principles and Insights*, McGraw-Hill (1968). For access, see this module's *Local Guide*.
3. G. R. Harrison, R. C. Lord, and J. R. Loofbourow, *Practical Spectroscopy*, Prentice-Hall (1948). For access, see this module's *Local Guide*.
4. R. A. Sawyer, *Experimental Spectroscopy*, Prentice-Hall (1946). For access, see this module's *Local Guide*.

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TRANSITIONS AND SPECTRAL ANALYSIS

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1. Introduction

Excited atoms and molecules drop down to their ground states, emitting photons characteristic of their source. Such photons constitute “fingerprints” of the atoms or molecules which emit them. Spectroscopy is the science of excitation of the atoms and molecules, detection of the photons subsequently emitted, and interpretation of the sources of the photons. Spectroscopy also includes detection of the absorption of photons in passing through a material. This unit examines how the photons are produced in atomic transitions, and how this process is put to practical use in spectroscopy.

2. Study Suggestions

Read:

M. Alonso and E. J. Finn, *Physics*, Addison-Wesley (1970).
For access, see this module's *Local Guide*.
Read Sections 25.1-25.4 and 27.7.
Examine Figure 27.28.

Then read:

1. I. M. Freeman, *Physics, Principles and Insights*, McGraw-Hill (1973), pp. 532-535 (Sec. 21.2-21.3). For access, see this module's *Local Guide*.
2. Harrison, Lord, and Loofbourow, *Practical Spectroscopy*, Prentice-Hall (1948), pp. 1-25 (skim sec. 1.14-1.20), pp. 27-29 (Sec. 2.1-2.2), and pp. 166-167. For access, see this module's *Local Guide*.
3. Sawyer, *Experimental Spectroscopy*, Prentice-Hall (1946), pp. 18-21, (Sec. 10 and 11), pp. 28-30 (Sec. 16 and first paragraph of Sec. 17). For access, see this module's *Local Guide*.

Acknowledgments

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